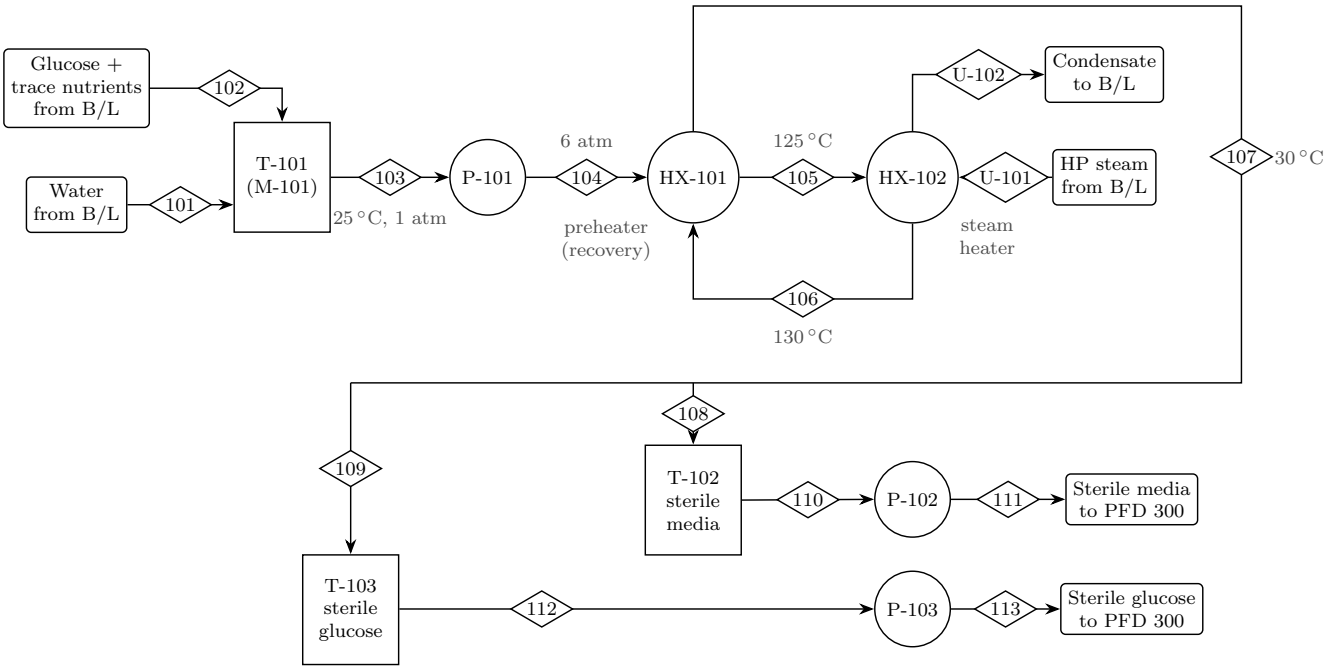
Heat Exchanger Network: Media and Glucose Sterilization

UBC CHBE capstone · Montbretin A production · PFD 100

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Process flow diagram



One sterilization train serves both feeds in separate campaigns: media (to T-102) and glucose (to T-103).

Nomenclature

$\dot{Q}$	heat-transfer rate (power) [kW]	A	heat-transfer area [m^2]
$\dot{m}$	mass flow rate [kg/s]	U	overall HT coefficient [$W/(m^2\,K)$]
c_p	specific heat (constant P) [$kJ/(kg\,K)$]	ΔT_{lm}	log-mean temperature difference [K]
ΔT	temperature difference [K]	F	LMTD correction factor [–]

Given

	Media campaign	**Glucose campaign**
Mass flow, $\dot{m}$	7196.8 kg/h	7103.3 kg/h
Composition	99.5 % water, 0.5 % media components	90.9 % water, 9.1 % glucose
Batch	14 394 kg (≈14 m^3) in 2 h	21 310 kg in ≈3 h
Temperatures	25 °C → 125 °C (HX-101) → 130 °C (HX-102) → 30 °C (HX-101 hot side)	
Steam	HP steam condensing at 150 °C	
c_p	4.18 kJ/(kg K) (water)	
U, steam heater	2000 Btu/(h ft^2 °F) = 11 356 W/(m^2 K)	
U, preheater	150 Btu/(h ft^2 °F) = 851.7 W/(m^2 K)	

The media flow basis: about 14 m^3 of media has to be processed in 2 h (lag phase, start-up), so 14 394/2 ≈

7197 kg/h. U values are preliminary-sizing heuristics from class notes, converted with 1 Btu/(h ft^2 °F) = 5.678 W/(m^2 K).

Equations

$$\dot{Q} = \dot{m}\, c_p\, \Delta T \qquad \text{(no phase change on the process side)} \tag{1}$$

$$\dot{Q} = U A\, \Delta T_{\text{lm}} F \;\Rightarrow\; A = \frac{\dot{Q}}{U\, \Delta T_{\text{lm}}} \qquad (F = 1) \tag{2}$$

$$\Delta T_{\text{lm}} = \frac{(T_{h,o} - T_{c,i}) - (T_{h,i} - T_{c,o})}{\ln\left(\dfrac{T_{h,o} - T_{c,i}}{T_{h,i} - T_{c,o}}\right)} \qquad \text{(counter-current)} \tag{3}$$

Heat leaves the hot fluid and enters the cold fluid: $|\dot{Q}_{\text{hot}}| = |\dot{Q}_{\text{cold}}|$.

Assumptions

- No heat losses; $F = 1$ (pure counter-current).
- Process streams stay liquid: P-101 pressurizes the feed to 6 atm.
- c_p of water for both media and glucose solution, constant over 25 °C to 130 °C.
- **Preheaters:** the hot and cold sides are the same stream, so both terminal temperature differences are 5 °C (130 − 125 and 30 − 25). The log-mean formula becomes 0/0, and its limit is $\Delta T_{\text{lm}} = 5$ °C. Counter-current flow is required: in co-current flow the cold outlet could never exceed the hot outlet.

Worked example

Log-mean temperature difference, steam heater (HX-102)

$$\Delta T_{\text{lm}} = \frac{(150 - 125) - (150 - 130)}{\ln\left(\dfrac{25}{20}\right)} = \frac{5}{0.2231} = 22.4\,^{\circ}\text{C}$$

Media campaign

$$\dot{Q}_{\text{steam}} = \left(7196.8\,\text{kg/h} \times \frac{1\,\text{h}}{3600\,\text{s}}\right)(4.18\,\text{kJ/(kg K)})(130\,^{\circ}\text{C} - 125\,^{\circ}\text{C}) = 41.8\,\text{kW}$$

$$\dot{Q}_{\text{pre}} = \left(7196.8\,\text{kg/h} \times \frac{1\,\text{h}}{3600\,\text{s}}\right)(4.18\,\text{kJ/(kg K)})(125\,^{\circ}\text{C} - 25\,^{\circ}\text{C}) = 835.6\,\text{kW}$$

$$A_{\text{steam}} = \frac{41.8 \times 10^3\,\text{W}}{(11\,356\,\text{W/(m}^2\,\text{K))}(22.4\,\text{K})} = \boxed{0.164\,\text{m}^2}$$

$$A_{\text{pre}} = \frac{835.6 \times 10^3\,\text{W}}{(851.7\,\text{W/(m}^2\,\text{K))}(5\,\text{K})} = \boxed{196.2\,\text{m}^2}$$

Glucose campaign

$$\dot{Q}_{\text{steam}} = \left(7103.3\,\text{kg/h} \times \frac{1\,\text{h}}{3600\,\text{s}}\right)(4.18\,\text{kJ/(kg K)})(130\,^\circ\text{C} - 125\,^\circ\text{C}) = 41.2\,\text{kW}$$

$$\dot{Q}_{\text{pre}} = \left(7103.3\,\text{kg/h} \times \frac{1\,\text{h}}{3600\,\text{s}}\right)(4.18\,\text{kJ/(kg K)})(125\,^\circ\text{C} - 25\,^\circ\text{C}) = 824.8\,\text{kW}$$

$$A_{\text{steam}} = \frac{41.2 \times 10^3\,\text{W}}{(11\,356\,\text{W/(m}^2\,\text{K))}(22.4\,\text{K})} = \boxed{0.162\,\text{m}^2}$$

$$A_{\text{pre}} = \frac{824.8 \times 10^3\,\text{W}}{(851.7\,\text{W/(m}^2\,\text{K))}(5\,\text{K})} = \boxed{193.7\,\text{m}^2}$$

Total network duty

$$\dot{Q}_{\text{total}} = 41.8 + 835.6 + 41.2 + 824.8 = 1743.4\,\text{kW} = \boxed{1.74\,\text{MW}}$$

Steam supplies only $41.8 + 41.2 = 83.0\,\text{kW}$ (4.8 %); the preheaters recover the other 95 % from the hot sterilized stream, and cooling back to 30 °C needs no cooling utility. The two campaigns run at different times, so the network handles about 0.88 MW at any one time.